



ENGINEERING MATHEMATICS - II

Dr. G. K. Jagatheswari

Department of Science and Humanities

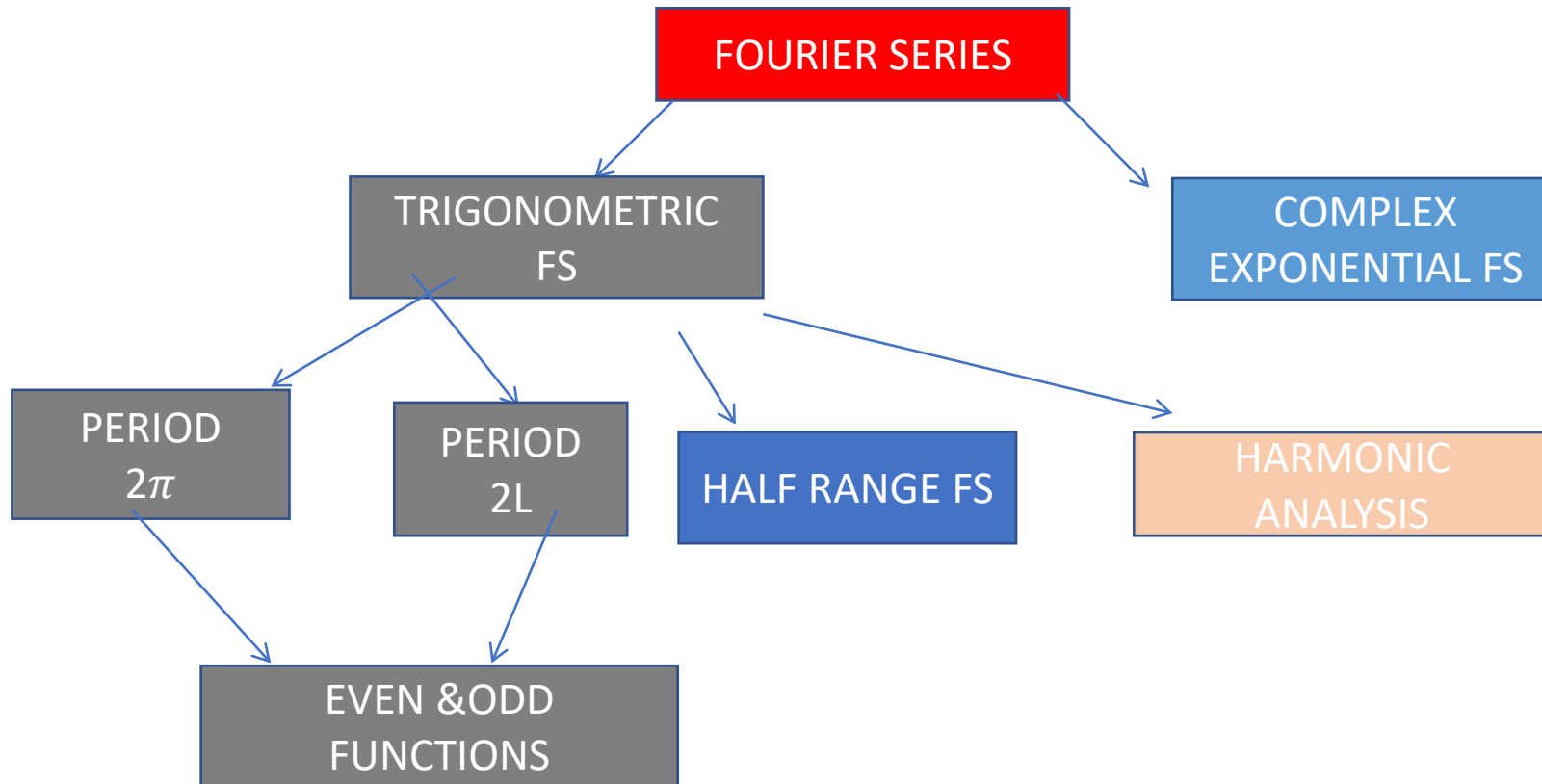
ENGINEERING MATHEMATICS - II

UNIT 5 : Fourier series

Session : 9

Sub Topic : Complex Form of the Fourier Series

Department of Science and Humanities



Complex form of the Fourier series



- ❖ The representation of signals over a certain time interval in terms of linear composition of exponential functions is called exponential Fourier series or complex form of the Fourier series.
- ❖ The complex form of Fourier series is algebraically simpler and more symmetric, when compared to Trigonometric Fourier series. Therefore, it is often used in physics and other sciences.
- ❖ The complex form of Fourier series is useful in problems on electric circuits having impressed periodic voltage.

Complex form of the Fourier series

Definition

The complex exponential form of the Fourier series of a function $f(x)$ defined in the interval $(-\pi, \pi)$ is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}$$

$$\text{where } c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx.$$

We know that, for a periodic function $f(x)$ of period 2π , defined over $(-\pi, \pi)$, the Fourier series expansion is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \dots \dots (1)$$

We know that $\cos nx = \frac{e^{inx} + e^{-inx}}{2}$ and

Complex form of the Fourier series

$$\sin nx = \frac{e^{inx} - e^{-inx}}{2i} = \frac{-ie^{inx} + ie^{-inx}}{2}$$

Therefore (1) becomes

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \frac{e^{inx} + e^{-inx}}{2} + b_n \frac{-ie^{inx} + ie^{-inx}}{2} \right) \\ &= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(\frac{a_n - ib_n}{2} \right) e^{inx} + \sum_{n=1}^{\infty} \left(\frac{a_n + ib_n}{2} \right) e^{-inx} \\ &= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(\frac{a_n - ib_n}{2} \right) e^{inx} + \sum_{n=-\infty}^{-1} \left(\frac{a_{-n} + ib_{-n}}{2} \right) e^{inx} \end{aligned}$$

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} \text{ where}$$

$$c_n = \begin{cases} \frac{a_0}{2}, & n = 0 \\ \frac{a_n - ib_n}{2}, & n = 1, 2, 3, \dots \\ \frac{a_{-n} + ib_{-n}}{2}, & n = -1, -2, -3, \dots \end{cases}$$

Now for n positive

$$c_n = \frac{a_n - ib_n}{2}$$

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} (\cos nx - i \sin nx) f(x) dx$$

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \dots\dots\dots(1)$$

Now for n negative

$$c_n = \frac{a_{-n} + ib_{-n}}{2}$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} (\cos(-nx) + i \sin(-nx)) f(x) dx$$

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \dots\dots\dots(2)$$

and for $n = 0$

$$c_n = \frac{a_0}{2}$$
$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-0ix} f(x) dx \quad \dots\dots\dots(3)$$

From (1), (2) and (3), we have

$$\therefore c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

Complex form of the Fourier series

Result:

The complex exponential form of the Fourier series of a function $f(x)$ defined on $(-l, l)$ having period $2l$ is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{in\pi x}{l}}$$

where $c_n = \frac{1}{2l} \int_{-l}^l f(x) e^{\frac{-in\pi x}{l}} dx.$

Complex form of the Fourier series - Problems

1. Find the complex exponential form of the Fourier Series of

$$f(x) = \cos ax, \quad -\pi < x < \pi.$$

Solution:

The complex exponential form of the Fourier series is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} \dots\dots\dots(1)$$

where

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-inx} \cos ax dx$$

Complex form of the Fourier series - Problems

$$c_n = \frac{1}{2\pi} \left[\frac{e^{-inx}}{a^2 - n^2} \{-in \cos ax + a \sin ax\} \right]_{-\pi}^{\pi}$$
$$= \frac{-in \cos a\pi}{2\pi(a^2 - n^2)} [e^{-in\pi} - e^{in\pi}] + \frac{a \sin a\pi}{2\pi(a^2 - n^2)} [e^{-in\pi} + e^{in\pi}]$$

$$c_n = \frac{a \sin a\pi}{2\pi(a^2 - n^2)} 2(-1)^n$$
$$\therefore c_n = \frac{a \sin a\pi}{\pi(a^2 - n^2)} (-1)^n$$

$f(x) = \cos ax = \sum_{-\infty}^{\infty} \frac{a \sin a\pi}{\pi(a^2 - n^2)} (-1)^n e^{inx}$ which is the required Fourier series.



THANK YOU

Dr. G K Jagatheswari

Department of Science & Humanities

jagatheswarigk@pes.edu